

MUST KNOW TOPICS BEFORE INTERVIEWS

Linux

Most Asked Linux Interview Questions for Freshers in MNCs

1. What happens when you type a command in terminal?
2. Difference between Process and Thread?
3. What is a Zombie Process?
4. What is a Daemon Process?
5. Difference between Hard Link and Soft Link?
6. What is an Inode in Linux?
7. Explain `chmod 755`.
8. Difference between `kill` and `kill -9`.
9. Difference between `grep`, `find`, and `locate`.
10. Difference between `>` and `>>`.
11. What is Pipe `|` in Linux?
12. What is Swap Memory?
13. What happens when RAM gets full?
14. Difference between TCP and UDP?
15. Difference between HTTP and HTTPS?
16. What is SSH?
17. What is Cron Job?
18. What is Environment Variable and PATH?
19. Explain `fork()` and `exec()`.
20. Your production server is down — how will you debug?

Commands

- `ls`
- `cd`
- `pwd`
- `grep`
- `find`
- `tail -f`
- `ps aux`
- `top`
- `chmod`
- `systemctl`
- `du`

Git

Most Asked Git Interview Questions for Freshers in MNCs

1. What is Git?
2. Difference between Git and GitHub?
3. What is a Repository?
4. Difference between `git pull` and `git fetch`?
5. Difference between `git merge` and `git rebase`?
6. What is a Branch in Git?
7. Why do we use Pull Requests?
8. What is Git Stash?
9. Difference between `git reset` and `git revert`?
10. What is Merge Conflict?
11. How do you resolve merge conflicts?
12. Difference between Local Repository and Remote Repository?
13. What is `.gitignore`?
14. Difference between `git clone` and `git fork`?
15. What is HEAD in Git?
16. Difference between `git add .` and `git add <file>`?
17. What is Detached HEAD state?
18. Explain Git workflow in teams.
19. How do you undo a commit?
20. How do you debug when code works locally but fails after merge?

Commands

- `git status`
- `git log --oneline --graph --all`
- `git checkout -b feature-name`
- `git pull origin main`
- `git rebase main`
- `git stash` , `git stash pop`
- `git cherry-pick <commit-id>`
- `git reset --soft HEAD~1`
- `git revert <commit-id>`
- `git reflog`

C++ — Top 10 Must Asked Core Questions

1. Stack vs Heap memory
2. Difference between compile time and runtime polymorphism
3. What happens during compilation in C++?
4. What is virtual function and how vtable works?
5. Difference between shallow copy and deep copy
6. Copy constructor vs move constructor
7. What are smart pointers and why needed?
8. Difference between process and thread
9. What causes memory leak and segmentation fault?
10. Difference between malloc/free and new/delete

Java — Top 10 Must Asked Core Questions

1. Difference between JDK, JRE, JVM
2. How Java achieves platform independence?
3. Heap vs Stack memory in JVM
4. How Garbage Collection works in Java?
5. Difference between == and equals()
6. Difference between HashMap and ConcurrentHashMap
7. Process vs Thread in Java
8. What is synchronization and deadlock?
9. What happens when a Java program runs internally?
10. Difference between String, StringBuilder, and StringBuffer

Python — Top 10 Must Asked Core Questions

1. What is GIL (Global Interpreter Lock)?
2. Difference between multithreading and multiprocessing
3. Mutable vs Immutable objects
4. Difference between is and ==
5. Deep copy vs shallow copy
6. How Python memory management works?
7. What happens when Python code executes internally?
8. List vs Tuple
9. What are generators and why are they memory efficient?
10. Difference between interpreter and compiler in Python

LeetCode Design Problems

- [LRU Cache](#)
- [LFU Cache](#)
- [Insert Delete GetRandom O\(1\)](#)
- [Design Twitter](#)
- [Min Stack](#)
- [Design Underground System](#)
- [Design Browser History](#)
- [Design HashMap](#)
- [Design Hit Counter](#)
- [Design Circular Queue](#)
- [Implement Trie \(Prefix Tree\)](#)
- [Design Add and Search Words Data Structure](#)
- [Time Based Key-Value Store](#)
- [All O`one Data Structure](#)
- [Design File System](#)

SQL

Top 10 Most Asked SQL Query Questions

1. Find nth highest salary using SQL
2. Find the 2nd highest salary from Employee table
3. Find duplicate records in a table
4. Delete duplicate rows from a table
5. Find highest salary department-wise
6. Find employees earning more than department average
7. Find employees with same salary
8. Find manager name along with employee name using self join
9. Find departments having more than 5 employees
10. Find customers who never placed orders

Top 10 Must Know SQL Interview Questions

1. Difference between DELETE, DROP, and TRUNCATE
2. Difference between WHERE and HAVING
3. Difference between INNER JOIN and LEFT JOIN
4. What is normalization and its types?
5. Difference between PRIMARY KEY and FOREIGN KEY
6. What is indexing and why is it used?
7. Difference between UNION and UNION ALL
8. What are ACID properties in SQL?
9. Difference between clustered and non-clustered index
10. What is subquery vs correlated subquery?

Top 10 Most Important Cloud Topics

- IaaS vs PaaS vs SaaS
- Virtual Machines vs Containers
- Cloud Computing Basics
- AWS Core Services (**EC2, S3, RDS, Lambda**) (Basic Understanding about Why it is Used)
- Load Balancer and Auto Scaling
- CI/CD Pipeline Basics
- Signed Url / Presigned Url
- Networking Basics (DNS, HTTP/HTTPS, VPC)
- Microservices and Distributed Systems Basics